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# **That Crisp, Clean Beer You Love Has Long Depended on a Strange, Very Old Ingredient. Brewers Are Dropping It in the Name of Progress.**

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# That Crisp, Clean Beer You Love Has Long Depended on a Strange, Very Old Ingredient. Brewers Are Dropping It in the Name of Progress.

As I listened to the calm cadence of Morgan Freeman's voice narrating yet another cataclysmic extinction event in Netflix's recent miniseries [The Dinosaur](#)s, my mind was necessarily on the Jurassic period. But rather than focusing on the famous land-before-time megafauna and their demise at the hands of the 2-million-year rainstorm of the [Carnian pluvial episode](#), I was admiring the impact of the prehistoric microbes that were their contemporaries. The Steven Spielberg-produced simulation had inconsiderately left them off-screen, but the light Pilsner I nursed had likely been poured over their fossilized remains.

Diatoms—a variety of tiny, single-celled algae that first emerged during the age of the dinosaurs—are a key component of the modern brewing industry, and a not-insignificant part of why your favorite beer looks and tastes the way it does. But that may be about to change. Clouds are forming in beer across the world, and brewers are increasingly abandoning the ancient algae that can clear them up.

Perhaps you know how beer is made, or you've heard about German beer-purity laws, *Reinheitsgebot*, which allow brewers to use only water, barley, hops, and yeast. It is less common knowledge, though, that much of the beer we are drinking (even that pure German stuff) filters through the fossilized remains of the prehistoric plankton that lived alongside our favorite dinos. Which is why even the casual beer drinker should take note of a white powder called diatomaceous earth. It has become one of the most common filtration substrates in industrial beer-making.

"It has a lot of characteristics that make it applicable in filtration," explains Roland Pahl, beer production expert at [Pall Corporation](#). "It has a big inner surface—the particles are shaped very bizarrely. The reason for that is that DE is basically fossilized algae. In ancient seas and ancient water reserves, algae would grow, die off, form a sediment on the ground, and fossilize over millions of years. It kept that shape intact but changed the structure to silicon dioxide, so it is almost inert."

Every year, breweries across the world collectively buy hundreds of thousands of tons of fossilized diatoms from mining companies in order to filter their beers. The global market for DE was valued at [\\$1.45 billion in 2025](#) (though many other sectors, including the dairy, edible oil, and petroleum industries, use it for filtration). The U.S. is the leading global producer of the substance, and about 60 percent of all DE ends up being used for filtration. But when you ask around in the industry today, it sounds as if Morgan Freeman is narrating another hapless species on the brink of extinction. The beer market is slowly turning its back on its fossilized fodder for good.

That doesn't mean, however, that modern consumers are about to give up their filtered suds. Up to [92 percent](#) of all beer consumed across the globe today is lager, that crisp, light, clear style we recognize in Corona, Heineken, Budweiser, and Coors, among others. Filtration is key to making every bottle, every pint, and every keg so clean and crushable.

"There's that old saying 'People drink with their eyes,' " says Chuck Skypeck, technical brewing projects director at the [Brewers Association](#). "The image of the crystal-clear beer with a nice head of foam on top is so inviting. A bacterial infection in a beverage is going to produce cloudiness, so

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there is probably something ingrained in our consciousness to view cloudiness with a doubtful eye.”

Drinkers should not take beer clarity for granted. A clear, crystalline Corona bottle would hardly be attractive if it were filled with a gruel-like mash of malt and water or the resulting unfiltered wort, each of which precedes beer in the brewing process. Fermented drinks like beer and wine are intentionally clarified once alcohol has been produced by yeast. There are both technical and psychological motivations for that clarity.

Pahl explains: “As soon as people know that a product can be clear, when they see it’s hazy, they suspect spoilage. It is easy for people to recognize turbidity.” Brewers who want to send consistent, stable beers out into the market are bound to rely on some filtration technique prior to bottling their product. It helps keep the beer tasting fresh. And they have tried their hand at many strange filters over the years.

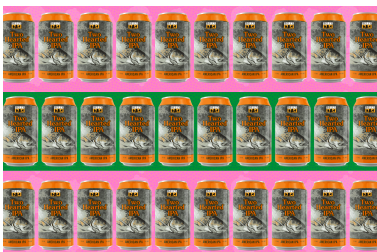
“In the late 17<sup>th</sup> and into the 18<sup>th</sup> century, brewers were trying numerous techniques and substances to clarify their beers, even dark and heavier beers like the porter,” Malcolm Purinton, a food historian and author of [Globalization in a Glass: The Rise of Pilsner Beer Through Technology, Taste and Empire](#), wrote me via email. “Among the options that were attempted were the use of flannel strainers attached to the exit pipes from fermenting vessels and a, thankfully brief, attempt at using asbestos as a filter bed.”

Shortly thereafter, a substance known as isinglass would become the norm for English brewers. Appearing in the swim bladder of fish, especially sturgeon, isinglass makes yeast precipitate out of the beer, clarifying it. Major breweries were still filtering their beer with these gelatinous fish organs until relatively recently—Guinness finally altered [its process in 2015](#) to make its dry stout vegan.

“By the end of the 18<sup>th</sup> century and early 19<sup>th</sup> century we see the additional use of *kieselguhr*, or diatomaceous earth as a filter bed,” Purinton continues. “Perlite also came to be used as well—siliceous glass made up of mostly aluminum silicate and volcanic in origin.”

For industrial brewers, though, DE has been the only viable option to filter hundreds of millions of liters of beer annually. “For the last four to five decades,” Pahl says, “DE filtration has really dominated the market. The major reason for filtration is stability. If you would leave the natural yeast in the product, the yeast would die off. And yeast does a nasty thing called autolysis that creates a lot of unpleasant flavor loads and other impacts on the quality—it changes the color of the product, it destroys foam, it enhances oxidation reaction.”

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Under a microscope, DE looks like a chalky, primordial Chex Mix. Brewers rely on the nooks and crannies and pores that once existed on prehistoric single-cell organisms to capture their own single-cell yeast alongside other yucky brewing sediment. The problem is that this kind of “dead-end” filtration clogs up DE’s pores. (And no one, not even an extinct microbe, wants clogged pores.) The stuff cannot be used again, meaning that it causes a lot of waste. The fossil filter also presents some major issues for breweries, including increased water usage, a relatively high CO<sub>2</sub> footprint, and generation of hundreds of tons of waste. In an age when breweries are under heightened scrutiny from educated, discerning consumers, sustainability is top of mind. Accordingly, consumers can expect to start drinking beers that rely on other, more renewable filtration techniques.

Pahl, whose company has moved away from producing and selling DE equipment to breweries, notes its drawbacks. “You cannot practically reuse it after it has been used in filtration,” he says. “Big breweries produce millions of hectoliters of beer a year—so hundreds of millions of liters. That makes the use of DE in the hundreds of tons. And once it has been used, it has been soaking up the product. It makes it wet, and it’s holding all the organics that you want to filter it out, so it multiplies its mass by a factor of three. A hundred tons going in makes 300 tons of waste going out.”

Where brewers did not previously have other options, technology is catching up to what is literally the dinosaur technique of the beer industry. Cartridge filters, smaller-scale centrifuges, and cross-flow membranes (which sweep debris away during filtration rather than clogging a filter with particles) are all increasingly common options for craft brewers. “I kind of look at DE as the grandfather filter medium that is starting to phase out,” explains Skypeck, of the Brewers Association. “It’s fairly difficult to work with, and it’s time-consuming. Working with that powder, it’s messy and it’s not easy to dispose of. You don’t want to put it down the drain.” Meanwhile, Pahl tells me, “cross-flow membrane filtration is, at this point in time, unanimously seen as the only alternative to DE filtration for large-scale breweries.” The wine industry has already largely moved away from diatoms, and beer is quickly following suit.

One final factor is hastening the fossil’s fall from favor—the rise of hazy beer. Across the craft-brewing industry, brewers are forgoing filtration altogether. New England IPAs, German Hefeweizens, and Kellerbiers are all unfiltered and hazy by design. These are beers still cloudy with the residual particles of the brewing process, and they generally deliver a full mouthfeel and a soft, equally cloudy texture on the tongue.

This haze is not so rampant as to be a [Pliensbachian-Toarcian Extinction Event](#)—level threat to the already-dead diatoms, but there’s no question that brewers are turning away from the Jurassic filtration substrate for lots of good reasons. It’s heavy, messy, water-intensive, time-consuming, and single-use. Well, it’s single-use as far as brewers are concerned. We cannot be sure what those same silica diatom skeletons will be doing in another 100 million years. We thank them, though, for their service to crisp, clear beer everywhere.

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